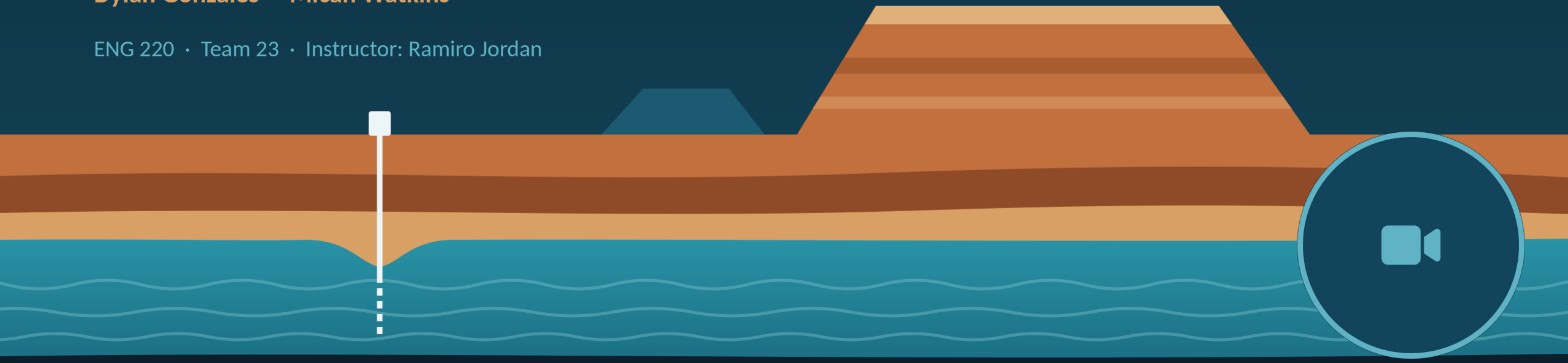
Beneath the Mesa

Assessing Aquifer, Well, and Acequia Water Resources Across Sandoval County, New Mexico

Petar Djordjevic (PM) · **Meckrem Yousef** · **James Wanueka**
Dylan Gonzales · **Micah Watkins**

ENG 220 · Team 23 · Instructor: Ramiro Jordan



Background

- Where Sandoval County's water comes from: wells, rivers, and acequias
- Why the county spans several aquifer systems
- The concern: declining groundwater near Cuba
- What this project set out to check



Figure caption and source

Research questions

MAIN QUESTION

Write the project's main question in one sentence

1. **Groundwater levels**
   Write the question in one sentence, using the wording from the proposal.

2. **Cuba's aquifers**
   Write the question in one sentence, using the wording from the proposal.

3. **Water quality and acequias**
   Write the question in one sentence, using the wording from the proposal.

Data

Groundwater levels

Source, years covered, and number of sites

Aquifer tests

Source, years covered, and number of sites

Water quality

Source, years covered, and number of sites

Acequias

Source, years covered, and number of sites

Full source links are on the last slide and in the project documentation.



Methods

Collect

Dylan Gonzales

One or two lines on what this step did

Clean

James Wanueka

One or two lines on what this step did

Analyze

Micah Watkins

One or two lines on what this step did

Present

Meckrem Yousef

One or two lines on what this step did

Tools Python · pandas · Plotly · Streamlit · GitHub

Results

Chart, map, or app screenshot

Copy this slide for each result

Figure caption and source

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Key number and what it measures

What this shows

Two or three sentences on the main pattern in the chart.

Findings

What the data show

- Measured results, with numbers

What they suggest

- Likely explanations, stated as possibilities

Too limited to tell

- Questions the data can't answer yet

Bottom line
One sentence that answers the main question



Limitations

- Areas with few or no monitoring wells
- Short or irregular water-level records
- Data the project couldn't get
- What a stronger study would need

Data coverage map

Where monitoring is dense, sparse, or absent

Figure caption and source

Explore the data, code, and app

GitHub repository
github.com/pdjordj11/eng_220_aquifer_project

Streamlit app
eng220aquiferproject.streamlit.app

Documentation
pdjordj11.github.io/eng_220_aquifer_project

Datasets
github.com/pdjordj11/eng_220_aquifer_project/tree/main/data

